

LOCALIZATION IN THE COMMUTANT, STAGE 17:

a study of “An isomorphism of the free group factors”.

The argument, its five constructs, its essence,
and what it changes for the programme

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Abstract

The paper (OpenAI, 23 September 2026) proves $L(\mathbb{F}_n) \cong L(\mathbb{F}_{n+1})$ for every $n \geq 3$. With Dykema’s amplification formula this gives $L(\mathbb{F}_2) \cong L(\mathbb{F}_3)$, and with the Dykema–Rădulescu alternative it shows that all interpolated free group factors are isomorphic, with fundamental group $\mathbb{R}_{>0}$. Free entropy dimensions are then not invariants of the generated von Neumann algebra.

- *Reading.* We reconstruct the proof in five moves and check each step’s logic.
- *Checks.* We test every computable lemma: the Fox cocycle and the prefix identity exactly; the free-Poisson spectral limit, the cutoff inversion and the free-sum norm bound in random-matrix models; and the mechanism of the key flow (letters barely move while a long word rotates by the extra generator).
- *Verdict.* We find no gap.

The essence, in one sentence: inside $L(\mathbb{F}_{n+1})$ there are trace-preserving automorphisms fixing the extra generator C that move each of the other n generators by an arbitrarily small amount in operator norm, while rotating one very long word in them by almost exactly C . Iterated with error budgets chosen after each step, they produce a norm-convergent sequence of free Haar n -tuples whose limit swallows C , and therefore generates everything. Four mechanisms make it work:

- a rich family of explicit automorphisms, the Fox-cocycle flows;
- coherent $\sqrt{m}$ amplification of tiny per-letter rotations along words with free prefixes;
- freeness turning ℓ^2 smallness into operator-norm smallness;
- the asymmetry between distribution, which is closed under norm limits, and generation, which is not.

We end with what this changes for our programme. Its central cocycle is literally our stage-15 error cocycle, and its adaptive iteration is a precise template for the instance-coupled dynamics the user has been asking for.

Addendum (Section 9). We confirm the companion study’s deductions and sharpen them:

- a relative absorption $P * L(\mathbb{Z}) \cong P * L(\mathbb{F}_{k+1})$ fixing P ;
- weak nullness of the discarded completions, which form a free Haar unitary in M^ω ;
- a hard-edge steering law, $\|T_m h - \delta_e\| \|h\| \geq (1 + o(1))/(2\sqrt{m})$, which forces $m = \Theta(\varepsilon^{-4})$ on the paper’s route against the universal $2/\varepsilon - 1$;
- the Fox flows as cocycle self-conjugacies of the free Bernoulli shift $M = (*_\Gamma L(\mathbb{Z})) \rtimes \Gamma$.

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1 Summary and verdict

What is proved.

- For $n \geq 3$, $L(\mathbb{F}_{n+1})$ contains a freely generating Haar n -tuple: n unitaries whose nonidentity reduced words all have trace zero and which generate the whole factor. So $L(\mathbb{F}_n) \cong L(\mathbb{F}_{n+1})$.

- Amplifying $L(\mathbb{F}_3) \cong L(\mathbb{F}_5)$ by $\sqrt{2}$ gives $L(\mathbb{F}_2) \cong L(\mathbb{F}_3)$.
- Hence all interpolated free group factors $L(\mathbb{F}_r)$, $1 < r \leq \infty$, are isomorphic, and their fundamental group is $\mathbb{R}_{>0}$.
- The free entropy dimensions $\delta, \delta_0, \delta^*, \delta^\star$ take every integer value $n \geq 2$ on finite generating tuples of $L(\mathbb{F}_2)$.

How. The construction has five moves, each elementary given freeness (Section 3).

1. Reduce isomorphism to finding a freely generating Haar tuple.
2. Take the logarithm $S = \arg C$ of the extra generator and its conjugates $S_g = A_g S A_g^*$. They form a free family indexed by $\mathbb{F}_n$.
3. Build polynomial flows $A'_j = i s(h_j) A_j$ with $s(k) = \sum_g k(g) A_g S A_g^*$, where the coefficients form a Fox cocycle.
 - The flows are trace-preserving, fix C , and move each letter by at most $3\pi \|h_j\|_{\ell^2}$.
 - They move a word A_w to within $3\pi \|D_w - \delta_e\|_{\ell^2}$ of $e^{iS} A_w$.
4. Make h tiny while $D_w \approx \delta_e$. With m free prefixes, $D_w = T_m h$ where $T_m T_m^*/m$ tends to the free Poisson law, which has no atom at 0. Truncated inversion gives $\|h\|_{\ell^2}^2 \approx 2/(\pi m \sqrt{d})$ with error $\approx (2/\pi)\sqrt{d}$.
5. Absorb and iterate. Compose with the basis change $C \mapsto C A_w$, so the moved n -tuple contains a word close to C . Repeat with budgets chosen after each word is known. The n -tuples converge in norm, keep the free Haar law, and their limit generates.

Verdict. We checked every step's logic and every computable lemma, and found no gap (Section 6). The weight-bearing steps are the trace-preservation argument (a first variation at a free point plus real-analytic continuation) and the spectral inversion (free Poisson without atom). Both hold. The external inputs are classical: the amplification formula and the alternative.

For us. The Fox cocycle $D_{gb} = D_g + \lambda(g) D_b$ is the same object as our stage-15 error cocycle. The paper's analytic heart, prefix transports by free unitaries adding like a circular element, is the mechanism behind our measured sum rule. Its adaptive iteration, where each tolerance is fixed only after the previous step's output is seen, is a rigorous template for the weight-coupled, dynamically adapted estimator we have been seeking (Section 8).

2 The statement and the classical reduction

For a discrete group G , $L(G) = \lambda(G)''$ on $\ell^2(G)$ with trace $\tau(x) = \langle x \delta_e, \delta_e \rangle$. The free group factor problem, attributed to Kadison, asks whether $L(\mathbb{F}_m) \cong L(\mathbb{F}_n)$ for $m \neq n$. Three classical facts frame it.

- *Interpolated factors and amplification* (Dykema, Rădulescu). There is a family $N_s = L(\mathbb{F}_s)$, $s > 1$, with

$$N_s^t \cong N_{1+(s-1)t^{-2}} \quad (s > 1, t > 0),$$

where Q^t is the corner of $Q \otimes B(\ell^2)$ of trace t .

- *The alternative* (Dykema; Rădulescu with $s = \infty$). Either all N_s , $1 < s \leq \infty$, are isomorphic, or they are pairwise non-isomorphic.
- *Haar tuples* (Lemma 2.2 of the paper). If unitaries $U_1, \dots, U_d$ generate M and every nonidentity reduced word has trace zero, then $\lambda(x_j) \mapsto U_j$ extends to a trace-preserving isomorphism $L(\mathbb{F}_d) \rightarrow M$. The proof is that the words form an orthonormal basis of $L^2(M)$.

So it is enough to find, inside $L(\mathbb{F}_{n+1})$, n unitaries with free Haar word distribution that generate it.

step	isomorphism	source
Theorem 1.1 at $n = 3, 4$	$N_3 \cong N_4 \cong N_5$	the construction
amplify by $\sqrt{2}$	$N_3^{\sqrt{2}} \cong N_5^{\sqrt{2}}$	functoriality
amplification formula	$N_3^{\sqrt{2}} = N_{1+2/2} = N_2$, $N_5^{\sqrt{2}} = N_{1+4/2} = N_3$	Dykema
conclusion	$L(\mathbb{F}_2) \cong L(\mathbb{F}_3)$	
alternative	all N_r isomorphic, $\mathcal{F}(N_r) = \mathbb{R}_{>0}$	Dykema–Rădulescu

The construction needs $n \geq 3$ because its prefix words use three letters, $a = x_1$ and $b_j = x_2^j x_3 x_2^{-j}$. Rank 2 comes only through amplification.

3 The construction in five moves

Fix $n \geq 3$ and $M = L(\mathbb{F}_{n+1})$ with canonical generators $(A_1, \dots, A_n, C)$, a freely generating Haar $(n+1)$ -tuple. Write $\Gamma = \mathbb{F}_n = \langle x_1, \dots, x_n \rangle$, and A_g for the word $g \in \Gamma$ evaluated in the first n letters.

3.1 Move 1: the bounded logarithm and its free conjugate family

Let $S = \arg C \in W^*(C)$, with values in $(-\pi, \pi]$. Then $S = S^*$, $\|S\| \leq \pi$, $\tau(S) = 0$, $e^{iS} = C$ and $W^*(S) = W^*(C)$. For $g \in \Gamma$ put $S_g = A_g S A_g^*$.

- *The conjugates are free* (Lemma 2.3). The algebras $A_g W^*(C) A_g^*$, $g \in \Gamma$, form a free family: cyclicity turns any alternating product into an alternating product of centred elements of $W^*(A)$ and $W^*(C)$. The free group indexes a free family of copies of S .
- *Group labels* (Lemma 2.4). Give a monomial in $A_j^{\pm 1}$ and S the product of its group letters as its *label*, ignoring S . Nonidentity labels have trace 0. Identity labels are polynomials in the S_g , obtained by moving letters right with $A_g S = S_g A_g$.

3.2 Move 2: Fox-cocycle flows

Choose finitely supported real $h_1, \dots, h_n \in \ell^2(\Gamma)$ and extend them to the cocycle $D : \Gamma \rightarrow \ell^2(\Gamma)$,

$$D_{x_j} = h_j, \quad D_{gb} = D_g + \lambda(g) D_b.$$

This is Fox's free differential calculus: a word's value is the sum over its letters of their coefficients, shifted by the preceding prefix. Consider the polynomial ODE, with S held fixed,

$$A'_j(t) = i s_t(h_j) A_j(t), \quad s_t(k) = \sum_g k(g) A_g(t) S A_g(t)^*.$$

Each letter is rotated on the left by a self-adjoint element built from conjugates of S by its own words. The product rule and the cocycle give the *word velocity*

$$A'_w(t) = i s_t(D_w) A_w(t),$$

so the cocycle value D_w is the generator of a word's rotation.

Proposition 3.1 (the flow; paper's Proposition 3.1). *The ODE has a global unitary solution. It defines a one-parameter group β_t of trace-preserving normal automorphisms of M fixing S (hence C). Moreover*

$$\|\beta_t(A_j) - A_j\| \leq 3\pi|t| \|h_j\|_{\ell^2}, \quad \|\beta_t(A_w) - e^{itS} A_w\| \leq 3\pi|t| \|D_w - \delta_e\|_{\ell^2}.$$

Trace preservation is the heart. It is proved without knowing freeness at positive times.

1. On the universal $*$ -algebra $\mathcal{P}$ (unitary a_j , self-adjoint z), define the derivation $\delta z = 0$, $\delta a_j = i\sigma(h_j)a_j$. It respects the relations, and $\delta z_g = i[\sigma(D_g), z_g]$.
2. *First variation at the free point.* δ preserves labels, so nonidentity-label terms have trace 0. On a polynomial $F(z_g)$, the derivative of z_g is $i[Y_g, S_g]$ with $Y_g = \sum_{b \neq g} D_g(b)S_b$; the own term commutes. Because S_g is free from $B_g = W^*(S_b : b \neq g) \ni Y_g$, conjugating S_g by e^{iuY_g} preserves the joint law with B_g . The first variation is therefore zero, and $\tau \circ \text{ev}_0 \circ \delta = 0$ on all of $\mathcal{P}$.
3. *Analytic continuation.* Applied to $\delta^{r-1}p$, step 2 gives $\frac{dr}{dt}\tau(\text{ev}_t p)|_0 = \tau(\text{ev}_0 \delta^r p) = 0$ for every $r \geq 1$. Unitarity of the evolving letters gives the Taylor bound $\frac{1}{r!} \|\frac{dr}{dt}\text{ev}_t p\| \leq b_0 R^l (B_1 R^J (l+J))^r$, uniformly in t . So $t \mapsto \tau(\text{ev}_t p)$ is real-analytic with a uniform radius, hence constant on $\mathbb{R}$.
4. *Descent.* If $\text{ev}_0 p = 0$ then $\tau(\text{ev}_t(p^*p)) = 0$, so the map is well defined. Since $\|b\| = \lim \tau((b^*b)^k)^{1/2k}$ it is isometric. Reversing time gives surjectivity, and the GNS unitary gives the normal extension.

Norm control comes from the free-sum inequality $\|\sum z_j\| \leq 2(\sum \|z_j\|_2^2)^{1/2} + \max \|z_j\|$ for free centred self-adjoint z_j . It gives $\|s(k)\| \leq 3\pi\|k\|_{\ell^2}$, *independently of the support size*. Integrating the velocities gives the two bounds, since $s_t(D_w) - S = \beta_t(s(D_w - \delta_e))$.

3.3 Move 3: the coefficient problem, small letters and a big word

Take $h_1 = h$ and $h_j = 0$ for $j \geq 2$. Use $a = x_1$, $b_j = x_2^j x_3 x_2^{-j}$, $p_j = (ab_1) \cdots (ab_j)$ and $w_m = p_m a$.

- *Free prefixes* (Lemma 4.2). The $u_j = ab_j$ are free, and $p_j = u_1 \cdots u_j$ is a triangular basis change, so $p_1, \dots, p_m$ are free.
- *Prefix identity.* D vanishes on $\langle x_2, \dots, x_n \rangle$, and a occurs in w_m after the prefixes $p_0, \dots, p_m$. Hence

$$D_{w_m} = T_m h, \quad T_m = \sum_{j=0}^m \lambda(p_j).$$

- *Spectral limit* (Lemma 4.3). $Q_m = T_m T_m^*$ has $\tau(Q_m^r) = \text{Cat}_r(m)_r + O(m^{r-1})$ from non-crossing matchings. So the spectral law of Q_m/m tends to the free Poisson law $d\nu = \frac{1}{2\pi} \sqrt{(4-x)/x} dx$ on $(0, 4)$: $T_m/\sqrt{m}$ becomes a circular element. Crucially, $\nu(\{0\}) = 0$.

- *Cutoff inversion* (Lemma 4.1). $h = T_m^* R_m \delta_e$, with $R_m = Q_m^{-1} \mathbf{1}_{(dm, \infty)}(Q_m)$, gives

$$T_m h = E_m \delta_e, \quad \|T_m h - \delta_e\|^2 = \mu_m([0, d]) \rightarrow \nu([0, d]) \approx \frac{2}{\pi} \sqrt{d}, \quad \|h\|^2 = \tau(R_m) \leq \frac{1}{dm}.$$

Real symmetry and truncation make h real and finitely supported.

So a vector of ℓ^2 norm of order $m^{-1/2} d^{-1/4}$ produces a cocycle value within $\approx d^{1/4}$ of δ_e . The word is $\sim m^2$ letters long.

3.4 Move 4: one absorption step

Let $\beta = \beta_1$ for these coefficients, and let γ be the automorphism induced by the free-group basis change $(x_1, \dots, x_n, c) \mapsto (x_1, \dots, x_n, cw)$. Then $\gamma(A_j) = A_j$ and $\gamma(C) = CA_w$. Put $\alpha = \gamma^{-1} \circ \beta$. Exactly,

$$\alpha(A_j) - A_j = \gamma^{-1}(\beta(A_j) - A_j), \quad \alpha(A_w) - C = \gamma^{-1}(\beta(A_w) - CA_w),$$

so $\max_j \|\alpha(A_j) - A_j\| < \varepsilon$ and $\|\alpha(A_w) - C\| < \varepsilon$. The new n -tuple is ε -close to the old one, and one of its words is ε -close to the old C . The new completion is $C' = CA_w^*$.

3.5 Move 5: the adaptive limit

Fix a dense sequence (y_j) in $L^2(M)$. At stage k the choices are made in this order:

1. approximate $y_1, \dots, y_k$ to 2^{-k} by polynomials $F_{j,k}$ in the current $(n+1)$ -tuple;
2. choose $\varepsilon_k < \min(r_{k-1}/2, 2^{-k}/(1 + H_k))$, with H_k the L^2 -Lipschitz constant of the $F_{j,k}$;
3. absorb (Move 4), obtaining the word w_k ;
4. substitute $G_{j,k}(X) = F_{j,k}(X, X_{w_k})$, which are polynomials in the n -tuple alone;
5. only now choose the future budget $r_k \leq \min(r_{k-1}/2, 2^{-k}/(1 + L_k))$, where L_k reflects the length of w_k .

Then $\sum_{\ell > k} \varepsilon_\ell \leq r_k$, so the n -tuples converge in operator norm to $A^{(\infty)}$. Each word's trace is norm-continuous, so $A^{(\infty)}$ is free Haar. Moreover $\|G_{j,k}(A^{(\infty)}) - y_j\|_2 \leq 3 \cdot 2^{-k}$, so the word algebra of $A^{(\infty)}$ is L^2 -dense. The trace-preserving conditional expectation onto $W^*(A^{(\infty)})$ then fixes a dense set, so $W^*(A^{(\infty)}) = M$. The completions $C^{(k)}$ need not converge.

4 The essence: four mechanisms

4.1 Mechanism 1: a gauge field on words, and why it must be outer

The flow rotates each letter on the left by a self-adjoint element built from conjugates of S . A word then rotates by the *accumulated* element $s(D_w)$, the Fox cocycle evaluated on the word. This is a gauge field along words: D plays the connection, D_w its holonomy, and $e^{iS} = C$ is the target holonomy.

Inner automorphisms fixing C cannot do this. $W^*(C)$ is a maximal abelian subalgebra of M , so they are $\text{Ad } u$ with $u = e^{if(S)}$, and they send $A_j \mapsto e^{if(S)} e^{-if(S_{x_j})} A_j$. Those are coboundary coefficients, $D_g = (1 - \lambda(g))v$ up to the functional calculus. A coboundary with $D_{x_j} = 0$ for $j \geq 2$

forces v to be invariant under $x_2, \dots, x_n$, hence $v = 0$. The absorption therefore uses *non-trivial* cocycles in $H^1(\mathbb{F}_n; \ell^2 \mathbb{F}_n)$.

That group is non-zero because $\beta_1^{(2)}(\mathbb{F}_n) = n - 1 > 0$. There is an irony here. The first L^2 -Betti number, long hoped to be the invariant that *distinguishes* the free group factors, is exactly what supplies the cocycles that make them isomorphic.

4.2 Mechanism 2: coherent $\sqrt{m}$ amplification along free prefixes

For h concentrated near e and free prefixes far apart, $\|T_m h\|^2 \approx (m+1)\|h\|^2$. Prefix shifts are nearly orthogonal, so the word velocity grows like $\sqrt{m}$ times the letter velocity even without design: this is a free central limit, $T_m/\sqrt{m} \rightarrow$ circular. Design does the rest. The minimal-norm solution $h = T_m^*(T_m T_m^*)^{-1} \delta_e$, with the inverse truncated, hits δ_e at cost $\|h\|^2 = \tau(R_m) \approx 2/(\pi m \sqrt{d})$.

This works only because the free Poisson law has *no atom at zero*, so T_m^* has no kernel in the direction of δ_e . Letters are moved by $O(m^{-1/2} d^{-1/4})$ while a word of length $\sim m^2$ is rotated by C up to $O(d^{1/4})$.

4.3 Mechanism 3: freeness turns ℓ^2 into operator norm

For bounded *commuting* independent variables, $\|\sum k_g X_g\|_\infty = \sum |k_g| \|X_g\|_\infty$ is an ℓ^1 quantity, since all extremes can coincide. For free variables, $\|\sum k_g S_g\| \leq 2\|S\|_2 \|k\|_{\ell^2} + \|S\| \max |k|$, an ℓ^2 quantity: free variables cannot conspire. This is what lets a coefficient vector of huge support but tiny ℓ^2 norm move every letter by a tiny amount in operator norm. In a commutative setting the same h would move letters by $\sim \sqrt{\text{support}} \cdot \|h\|_{\ell^2}$, which is large.

Non-commutativity is the resource. The free family $\{S_g\}$ indexed by the free group is what makes small-but-spread vector fields small.

4.4 Mechanism 4: distribution is closed, generation is not

Having free Haar word distribution is a set of trace conditions on words, each continuous in operator norm, so it survives norm limits. Generating M is a density condition, which is neither open nor closed under norm limits. Each $A^{(k)}$ generates a proper subfactor with a free complement, yet the limit generates everything.

The adaptive budgets are what make generation survive. Each target y_j , once approximated by a polynomial in the current n -tuple, is protected by all later budgets, which are chosen only after the word length that the polynomial needs is known. It is a diagonal argument in which the complexity of the witnesses (word lengths) grows without bound while the moves shrink faster.

4.5 The whole argument in one line

$$\underbrace{\beta_1^{(2)}(\mathbb{F}_n) > 0}_{\text{outer Fox flows}} + \underbrace{\nu_{\text{FP}}(\{0\}) = 0}_{\text{small letters, big word}} + \underbrace{\text{free Khintchine}}_{\ell^2 \rightarrow \|\cdot\|} + \underbrace{\text{adaptive budgets}}_{\text{generation in the limit}} \implies L(\mathbb{F}_n) \cong L(\mathbb{F}_{n+1}).$$

5 Why it evaded thirty years, and what changes

5.1 The direction that failed

For thirty years the main line of attack sought an *invariant that counts generators*. Candidates were Voiculescu’s free entropy dimensions $(\delta, \delta_0, \delta^*, \delta^\star)$, Connes–Shlyakhtenko L^2 -Betti numbers of the algebra, and cost-like quantities. On canonical generators they all return n (or $n - 1$). The expert’s remark, that “we have been heading in the wrong direction”, reads naturally against this background. The paper shows that the count is a property of the *generating set*, not of the algebra:

- a free Haar n -tuple, with $\delta_0 = n$, generates $L(\mathbb{F}_{n+1})$;
- $L(\mathbb{F}_2)$ has finite generating tuples with $\delta = \delta_0 = \delta^* = \delta^\star = n$ for every $n \geq 2$ (Corollary 7.1).

The resolution came instead from *dynamics*: an explicit, rich family of automorphisms and an adaptive limit. That is a constructive object, not a numerical invariant.

5.2 What stays true, and what must change

- *Upper-bound theorems survive.* These are results that bound δ_0 for *all* generating sets, such as Voiculescu’s absence of Cartan subalgebras, Ge’s primeness, Jung’s strong 1-boundedness, and Hayes’ 1-bounded entropy. They are statements about every generating set, and remain valid. $L(\mathbb{F}_n)$ still has no Cartan subalgebra, is prime and strongly solid, and $h(L(\mathbb{F}_n)) = \infty$ for all n .
- *Generator-counting invariants must collapse.* Any quantity that is an invariant of the von Neumann algebra and was expected to equal n or $n - 1$ on $L(\mathbb{F}_n)$ must take a single value for all n . For example, the Connes–Shlyakhtenko first L^2 -Betti number of $L(\mathbb{F}_n)$ cannot equal $n - 1$ for every n .
- *Fundamental group.* $\mathcal{F}(L(\mathbb{F}_2)) = \mathbb{R}_{>0}$; every interpolated factor is isomorphic to every other, including $L(\mathbb{F}_\infty)$.
- *Generation is not stable under small perturbations.* Choosing r_0 small in the construction gives free Haar n -tuples *arbitrarily close in operator norm* to the canonical generators $\lambda(x_1), \dots, \lambda(x_n)$ of $L(\mathbb{F}_{n+1})$ that generate all of it. The canonical tuple generates a proper subfactor with a free complement. Generation is invisible to any norm-local count.

5.3 Relation to earlier techniques

- *Voiculescu’s cyclomorphy* exponentiates trace-preserving polynomial vector fields on free semi-circular generators. The paper redoes it for Haar unitaries with a fixed bounded logarithm, proving descent through the unitary relations from moments.
- *Guionnet–Shlyakhtenko free monotone transport* builds isomorphisms by changes of variables at a *fixed* rank. Here the rank changes through an infinite product of small automorphisms and a limiting generation argument.
- *Shlyakhtenko’s Fuchsian preprint* (September 2026) finds free complements by finite automorphisms plus limiting generation. The idea was in the air, and the small-cocycle absorption is the new ingredient.
- *Fox calculus* (1953) supplies the cocycle rule, and *Ricard–Xu* the free Khintchine decomposition.

6 Critical reading

We went through every step, looking for the places where a proof of this kind usually breaks.

step	what could fail, and why it does not
Haar-tuple identification (L. 2.2)	needs only orthonormality of words and density of their span;
free conjugates (L. 2.3)	both follow from the trace conditions and generation
labels (L. 2.4)	cyclic rotation turns $\tau(\prod A_{g_j} c_j A_{g_j}^*)$ into an alternating centred product of free algebras; correct
derivation on $\mathcal{P}$	the alternating-reduction induction is standard; nonidentity labels survive the removal of scalar parts
first variation (L. 3.2)	respects $a_j^* a_j = 1$ since $\sigma(h_j)$ is self-adjoint (h_j real); the cocycle rule follows from the product rule
higher derivatives analyticity (L. 3.4)	the own term commutes; $Y_g \in B_g$; conjugating one free variable by a unitary of its free complement preserves the joint law. Freeness is used <i>only at</i> $t = 0$
descent, isometry, surjectivity norm bound (L. 2.5)	$\tau \circ \text{ev}_0 \circ \delta = 0$ holds on all of $\mathcal{P}$, hence on $\delta^{r-1}p$ the Taylor bound is uniform in t because the evolved letters are unitary and S is fixed; constancy on $\mathbb{R}$ follows from real-analytic continuation
prefix identity (L. 4.2)	faithfulness of τ ; $\ b\ = \lim \tau((b^*b)^k)^{1/2k}$; uniqueness of the reversed flow
free Poisson limit (L. 4.3)	creation, annihilation and diagonal decomposition on the free product L^2 ; constant 3π (we measure $2\pi/\sqrt{3}$ at the free CLT edge)
cutoff inversion (L. 4.1)	checked exactly by computer (Section 7)
absorption (P. 5.1)	non-crossing matching count plus tightness and moment determinacy on $[0, 4]$; we observe it numerically
adaptive limit (Thm 1.1)	order of choices $d \rightarrow m \rightarrow$ finite truncation; real symmetry via conjugation commuting with real functional calculus
	exact algebra with $\gamma(C) = CA_w$; the error transfers isometrically
	ε_k chosen before w_k, r_k after; $\sum_{\ell > k} \varepsilon_\ell \leq r_k$; L^2 -density plus conditional expectation gives generation

Further remarks.

- *Rates.* With the actual spectral asymptotics, $\|h\|^2 \approx 2/(\pi m \sqrt{d})$ and $\text{error}^2 \approx (2/\pi)\sqrt{d}$. Accuracy η needs $d \sim (\pi\eta^2/2)^2$ and $m \sim 4/(\pi^2\eta^4)$, with word length $\sim m^2 \sim \eta^{-8}$. That is polynomial blow-up, irrelevant for existence and a mild cost for any numerical realization.
- *Constructivity.* Each stage is explicit, with an explicit word, coefficient vector and flow. The limit is an infinite product, and the isomorphism has no closed form.
- *External inputs.* Only Dykema's amplification formula is used for Theorem 1.2, and only the alternative for Corollary 6.1. Both are classical.
- *What a referee will read hardest.* The first-variation argument, and the claim that real-analytic continuation needs no freeness at $t > 0$. We find both correct. The authors' care in separating formal identities on $\mathcal{P}$ from evaluated ones is what makes the second step legitimate.

7 Checks

The script `scripts/verify_ffac.py` runs in a few minutes. Its transcript is `notes/stage17/verify_ffac.txt`.

statement	result
prefix identity $D_{w_m} = T_m h$ (L. 4.2)	exact (difference 0) for $m = 1, 3, 5$, with word lengths 5, 19, 41
cocycle identity $D_{uv} = D_u + \lambda(u)D_v$	exact on 200/200 random pairs
free prefixes	936 reduced words of length ≤ 4 in p_1, p_2, p_3 : all distinct, none trivial
free Poisson limit (L. 4.3), $N = 1200$	moments of Q_m/m at $m = 128$: 1.007, 2.021, 5.060, 14.18 (Catalan 1, 2, 5, 14); $\mu([0, 0.05]) = 0.1417$ vs $\nu = 0.1421$; top eigenvalue 3.95 (edge 4)
cutoff inversion (L. 4.1)	$m = 128$, $d = 0.02$: $\ h\ ^2 = 0.031$ (bound 0.39), $\text{error}^2 = 0.089$ ($\nu([0, d]) = 0.090$)
rates	$\eta = 0.3$: $m \approx 50$; $\eta = 0.1$: $m \approx 4 \times 10^3$, word length $\approx 1.6 \times 10^7$
free-sum bound (L. 2.5), $N = 600$	spread coefficients: 20.1 against bound 52.6 and free-CLT edge 20.2; one large coefficient: 30.6 against 81.9
flow mechanism (P. 3.1), $N = 160$	$m = 16$: letter velocity 0.93, word deviation $\ s(D_w) - S\ = 1.38$; $m = 64$: 0.74 and 1.02; $\ S\ = 3.13$; within the 3π bounds
amplification arithmetic steering law (Sec. 9), limit law	$N_3^{\sqrt{2}} = N_2$, $N_5^{\sqrt{2}} = N_3$ $\sqrt{m} \text{err } \ h\ $: Tikhonov 0.475, 0.492, 0.4975, 0.4992 at $d = 10^{-2}, \dots, 10^{-5}$ ($\rightarrow \frac{1}{2}$ from below); cutoff 0.611, $\dots$, 0.6358 ($\rightarrow 2/\pi = 0.6366$)
steering law, finite m	$m = 128$, $d = 0.01$: Tikhonov 0.470, cutoff 0.606 (random-matrix spectra)
$\ C - A_w\ = 2$ (8.3), $N = 400$	2.0000; $\ C - A_w\ _2 = 1.4151$ ($\sqrt{2} = 1.4142$); largest spectral gap of C^*A_w on the circle 0.035

In a matrix model, conjugation fixes the identity matrix, so S must be centred and the trivial direction removed. In the free setting the conjugates of the centred S span a copy of $\ell^2(\Gamma)$, which has no such direction. The mechanism check shows the paper's trade-off: as m grows, letters move less while the word's rotation approaches the pure rotation by S . Convergence is slow, as the rates predict. In the steering law the $o(1)$ is negative at moderate d , because the free Poisson density lies below its edge asymptote $1/(\pi\sqrt{x})$.

8 What it means for our programme

8.1 The same cocycle

The paper's Fox cocycle and our stage-15 error cocycle are one structure:

$$D_{gb} = D_g + \lambda(g)D_b \quad \longleftrightarrow \quad \delta_l = T_l \delta_{l-1} + \beta_l, \quad \delta_L = \sum_l T_L \cdots T_{l+1} \beta_l.$$

In both, a word's value is the sum of local contributions, each transported by part of the word: the prefix (left regular action) in theirs, the downstream gated maps in ours. Their analytic heart

concerns the Gram of the transports. For free prefixes it tends to a circular element, with no atom at zero. Ours is what the measured sum rule reports: births from different layers add in quadrature, with transport factors 0.65–0.94.

- *Incoherent local contributions add in ℓ^2 .* The word grows like $\sqrt{m}$. This is our late, incoherent residual.
- *Designed contributions can be coherent.* The minimal-norm input $T^*(TT^*)^{-1} \cdot \text{target}$ reaches an $O(1)$ target with local inputs of size $m^{-1/2}$. For an estimator this is the adjoint (dual-weighted) design of corrections, and it is efficient exactly when the transport Gram has no small singular values in the target direction.

8.2 The same orthogonality

Free Khintchine, $\|\sum k_g S_g\| \lesssim \|k\|_{\ell^2}$, is the non-commutative form of our row-Mehler theorem: independence across rows makes errors add in ℓ^2 rather than ℓ^1 . In the paper this is a *resource*, since spread corrections are small. For us it is a *constraint*, since spread errors cannot be cancelled by a few global corrections (stage 15, Corollary 3.3).

8.3 The same lesson about counting

The paper shows that a generator count is a property of the generating set, not of the algebra; the invariant structure is dynamical. Our stage 16 measured the analogue for estimators. A representation in a fixed basis needs degree ~ 17 at depth 16, and the tractable representation co-evolves with the weights. In both cases the productive object is a *flow of frames* adapted to the instance, not a count.

8.4 A template for dynamic interdependence

The paper’s iteration is the cleanest rigorous example we know of a computation whose next step is fixed only after the previous output is seen. Each pass makes a small, structure-preserving move that serves a target, and the tolerance of every future move is set by the sensitivity of everything already achieved. The transferable design principle for an estimator has three parts:

- *Passes of small, exact moves.* A sequence of passes, each a small move of the internal frame that preserves the exact identities (homogeneity, Euler–Stein, moment positivity). This is the analogue of trace preservation.
- *Moves aimed through the adjoint.* Each move is directed at the currently unresolved part, the analogue of absorbing C .
- *Budgets set by sensitivity.* Each pass’s tolerance comes from the adjoint sensitivities of what earlier passes achieved, the analogue of r_k .

What does not transfer. Mechanism 3 is non-commutative. Our network’s state lives in a commutative probability space, where spread moves cost ℓ^1 , so the absorption phenomenon itself has no classical analogue. What transfers is the cocycle algebra, the spectral (Gram) analysis of transports, and the order of choices in the adaptive iteration.

9 Addendum: deductions from the companion proof study

The companion proof study (*Absorbing a free generator*, 13 pp) reconstructs the argument independently. It adds three deductions (its 8.1–8.3) and reads M as a crossed product (its §9). We confirm all four. We also sharpen three of them and add one quantitative law:

- a relative form of the theorem that fixes a subfactor pointwise;
- weak nullness of the completions;
- a hard-edge steering law that fixes the cost exponent of the paper’s route;
- the identification of the Fox flows as cocycle self-conjugacies of the free Bernoulli shift.

Two verification caveats from the study stand. We have not rerun the reported Lean formalization, and the quoted expert is not authenticated. Our verdict rests on our own reading.

9.1 Only one letter moves (study 8.1), and a relative absorption

In Move 3, $h_1 = h$ and $h_j = 0$ for $j \geq 2$, so $A'_j = 0$ for $j \geq 2$ and β fixes $A_2, \dots, A_n$. The basis change γ fixes every A_j , so α fixes $A_2, \dots, A_n$. Every stage of the adaptive limit does the same with the current first letter. The limit tuple is therefore $(B_1, A_2, \dots, A_n)$ with $\|B_1 - A_1\| < \delta$.

Corollary 9.1 (relative absorption). *Let $n \geq 3$, $P = W^*(A_2, \dots, A_n) \cong L(\mathbb{F}_{n-1})$ and $k \geq 1$. There is an isomorphism $P * L(\mathbb{Z}) \cong P * L(\mathbb{F}_{k+1})$ that is the identity on P and sends the generator of $L(\mathbb{Z})$ to within any prescribed δ of A_1 . In particular a fixed copy of $L(\mathbb{F}_{n-1})$ has free complements $W^*(B_1) \cong L(\mathbb{Z})$ (abelian) and $W^*(A_1, C) \cong L(\mathbb{F}_2)$ (a factor) in the same algebra. Free complements are not unique even up to isomorphism.*

Proof. For $k = 1$, apply the theorem to the n -tuple $(A_1, \dots, A_n)$ with completion C . Only A_1 moves, and the limit tuple is free Haar, so B_1 is free from P . For general k , start from $(A_1, \dots, A_n, C_1, \dots, C_k)$ and absorb $C_1, \dots, C_k$ in turn into the current first letter, using the tuple $(B_1^{(i)}, A_2, \dots, A_n, C_{i+2}, \dots, C_k)$ at step $i + 1$. The prefix words use only the first three letters, so $n \geq 3$ suffices, and the δ_i can be summed. $\square$

9.2 Strong but never uniform recovery (study 8.2), and an escaping completion

Let $\mathcal{A}_k = W^*(A^{(k)})$ and let E_k be its trace-preserving conditional expectation. The study proves $E_k \rightarrow I$ strongly on $L^2(M)$:

- $\|E_k y_j - y_j\|_2 \leq \|G_{j,k}(A^{(k)}) - y_j\|_2 \rightarrow 0$ on a dense set;
- $\|E_k\| \leq 1$ extends this to all of $L^2(M)$.

It also proves $\|I - E_k\| = 1$: $C^{(k)}$ is a Haar unitary free from $\mathcal{A}_k$, so $E_k C^{(k)} = \tau(C^{(k)}) = 0$ with $\|C^{(k)}\|_2 = 1$. We add:

Proposition 9.2. *$C^{(k)} \rightarrow 0$ weakly in $L^2(M)$. Hence the completions have no L^2 limit, where the paper’s “need not converge” leaves this open. In the ultrapower, $(C^{(k)})_\omega \in M^\omega$ is a Haar unitary free from M .*

Proof. For $y \in L^2(M)$, $|\langle C^{(k)}, y \rangle| = |\langle C^{(k)}, (I - E_k)y \rangle| \leq \|(I - E_k)y\|_2 \rightarrow 0$.

For freeness, take an alternating product of centred $y_i \in M$ and centred polynomials in $C^{(k)}$. Replace each y_i by the centred, norm-bounded $E_k(y_i)$. This changes the trace by at most $\sum_i \|y_i - E_k y_i\|_2 \prod_{j \neq i} \|x_j\| \rightarrow 0$, and the replaced product has trace 0 by freeness at stage k . $\square$

Popa showed that every separable II_1 factor has such a free Haar unitary in M^ω . Here the construction produces one explicitly, as the discarded completions. Recovery is target by target: every fixed observable is recovered, while a unit direction, the current completion, is always invisible and drifts off to infinity.

9.3 Word length (study 8.3), refined, and a steering law

C is free from $W^*(A)$, and A_w is a Haar unitary for $w \neq e$. So C^*A_w is Haar, its spectrum is the whole circle, and $\|C - A_w\| = 2$. The study telescopes this against ε -moves of the letters to get $|w| > 2/\varepsilon - 1$. By 8.1 only occurrences of $x_1^{\pm 1}$ cost anything.

Proposition 9.3 (universal bound). *If an absorption moves only the first letter, by less than ε , and $\|B_w - C\| < \varepsilon$, then $\ell_1(w) > 2/\varepsilon - 1$, where $\ell_1(w)$ is the number of occurrences of $x_1^{\pm 1}$ in w .*

In the construction $\ell_1(w_m) = m + 1$, while $|w_m| = m^2 + 3m + 1$: the letters $b_j = x_2^j x_3 x_2^{-j}$ are padding that keeps the prefixes free. How large must m be on the paper's route? The answer is set by the coefficient problem of Move 3.

Proposition 9.4 (steering law). *Let μ_m be the spectral measure of $Q_m = T_m T_m^*$ at δ_e . For every h and every $\lambda > 0$,*

$$\|T_m h - \delta_e\|^2 + \lambda \|h\|^2 \geq \int \frac{\lambda}{q + \lambda} d\mu_m(q),$$

with equality at the Tikhonov solution $h = T_m^(Q_m + \lambda)^{-1} \delta_e$. Take $\lambda = md$ with $1/m \ll d \ll 1$ and replace $\mu_m(m \cdot)$ by its free Poisson limit ν , whose density is $\sim 1/(\pi\sqrt{x})$ near 0. The right side is then $\sqrt{d}(1 + o(1))$. Optimising over d gives the Pareto frontier*

$$\|T_m h - \delta_e\|_{\ell^2} \cdot \|h\|_{\ell^2} \geq \frac{1 + o(1)}{2\sqrt{m}}.$$

Tikhonov attains it. The paper's spectral cutoff attains $2/(\pi\sqrt{m})$, within a factor $4/\pi$.

Proof. The inequality is the minimum of a quadratic, with value $\lambda \langle \delta_e, (Q_m + \lambda)^{-1} \delta_e \rangle$. Also $\int_0^\infty \frac{d}{x+d} \frac{dx}{\pi\sqrt{x}} = \sqrt{d}$. With $a = m\|h\|^2$, $\|T_m h - \delta_e\|^2 \geq \sup_d (\sqrt{d} - da) = 1/(4a)$.

For the cutoff, $\nu([0, d]) \approx \frac{2}{\pi}\sqrt{d}$ and $\int_d \frac{d\nu}{x} \approx \frac{2}{\pi\sqrt{d}}$, whose product is $4/\pi^2$. $\square$

Corollary 9.5 (cost of the Fox-prefix route). *Asking for letter moves and the word error both below ε , with the proven constant 3π , is sufficient, to leading order, once $m \geq 81\pi^4/(4\varepsilon^4)$. To first order, $\|s(k)\| \geq \|S\|_2 \|k\|_2 = (\pi/\sqrt{3})\|k\|_2$, so $m \gtrsim \pi^4/(36\varepsilon^4)$ is necessary on this route.*

Hence the route needs $m = \Theta(\varepsilon^{-4})$ moved letters and $|w| = \Theta(\varepsilon^{-8})$, against the universal $\ell_1(w) > 2/\varepsilon - 1$.

Remark 9.6 (the exponent is the hard edge). Suppose the Gram law near 0 is $\nu([0, x]) \sim cx^\alpha$ with $0 < \alpha < 1$. The same Legendre argument gives $m \gtrsim \|h\|^{-2} \text{err}^{-2(1-\alpha)/\alpha}$.

- A spectral gap costs nothing in accuracy.

- An atom at 0 makes the target unreachable.
- The free Poisson law has the square-root hard edge $\alpha = \frac{1}{2}$, which is what fixes the paper's ε^{-4} .

With free prefixes, shorter padding could shorten $|w|$ but cannot change the exponent in m . Whether a different absorption mechanism closes the gap between ε^{-1} and ε^{-4} is open.

9.4 The crossed product (study §9), and what the flows are in it

Let $\mathcal{N} = W^*(S_g : g \in \Gamma)$. By Lemma 2.3 it is the free product $*_{g \in \Gamma} W^*(S_g)$ of copies of $L(\mathbb{Z})$. The unitaries A_g normalise it, $\sigma_g := \text{Ad } A_g$ shifts the index, and by Lemma 2.4 $E_{\mathcal{N}}(A_g) = 0$ for $g \neq e$. Together with $C = e^{iS} \in \mathcal{N}$, this gives the study's

$$M = \mathcal{N} \rtimes_{\sigma} \Gamma \quad (\text{the free Bernoulli shift of } \Gamma = \mathbb{F}_n \text{ with base } L(\mathbb{Z})),$$

and $L^2(M) = \bigoplus_g L^2(\mathcal{N}) A_g$, graded by labels. The theorem then reads $(*_\Gamma L(\mathbb{Z})) \rtimes \Gamma \cong L(\Gamma)$: the crossed product absorbs its coefficient algebra into the group algebra.

Proposition 9.7 (Fox flows are cocycle self-conjugacies). *For all t , $\beta_t(\mathcal{N}) = \mathcal{N}$ and $\beta_t(A_g) = U_g(t) A_g$ with $U_g(t) \in \mathcal{U}(\mathcal{N})$. Moreover:*

- $U_{gb} = U_g \sigma_g(U_b)$, so $U(t)$ is a unitary 1-cocycle of the shift;
- $\theta_t = \beta_t|_{\mathcal{N}}$ satisfies $\theta_t \sigma_g \theta_t^{-1} = \text{Ad } U_g(t) \sigma_g$ and $\theta_t(S) = S$;
- $\frac{d}{dt} U_g(t)|_0 = i s(D_g)$.

So β_t preserves the grading. The Fox cocycle, pushed through the equivariant map $s : \ell^2 \Gamma \rightarrow \mathcal{N}$ (with $\sigma_g \circ s = s \circ \lambda(g)$), is the tangent of a unitary cocycle of the free Bernoulli shift.

Proof. The derivation δ preserves labels. So $\text{ev}_0(\delta^r a_j)$ lies in $\mathcal{N} A_j$, because a label- x_j monomial times A_j^* is a polynomial in the S_g by Lemma 2.4. The Taylor series converges in norm for $|t| < R$, with the uniform radius of Lemma 3.4. Since $\mathcal{N} A_j$ is norm closed, $\beta_t(A_j) \in \mathcal{N} A_j$ for $|t| < R$.

The cocycle rule follows from $\beta(A_g A_b) = U_g A_g U_b A_b$. Then $\beta_t(S_g) = U_g S_g U_g^* \in \mathcal{N}$, and with β_{-t} this gives $\beta_t(\mathcal{N}) = \mathcal{N}$. The group law extends everything to all t . The derivative is the word velocity of Move 2 at $t = 0$. $\square$

The basis change γ is the one move that breaks the grading. $\gamma(C) = C A_w$ lies in the w -graded piece $\mathcal{N} A_w$, so $\gamma(\mathcal{N}) \neq \mathcal{N}$. Each absorption step is therefore:

1. a continuous, grading-preserving cocycle move β , which steers the w -component of A_w to $\approx e^{iS} A_w = C A_w$;
2. a discrete regrading γ^{-1} by w .

The rank changes only through the discrete regradings, and each one costs $\ell_1(w) > 2/\varepsilon - 1$.

We agree with the study's caveats. β_t is not a modular flow, since τ is a trace and the modular group is trivial. D is a group cocycle, not a Dirac operator. An abstract isomorphism forgets the marked generators, and with them all geometric and computational data.

9.5 For the programme

- *A steering law for our transports (conjecture, to measure).* The stage-15 no-free-lunch for local corrections should have the quantitative form above. Cancelling a target at depth gap k with corrections of size $\|h\|$ should cost $\|h\|^{-2}\text{err}^{-2(1-\alpha_k)/\alpha_k}$, where α_k is the hard-edge exponent of the gated transport Gram $T_L \cdots T_{L-k+1}(\cdot)^*$ in the output-mean direction. Products of k free circular factors have Fuss–Catalan edges $\alpha = 1/(k+1)$, which would predict a cost err^{-2k} . Measuring α_k on net 0 from the stage-16 Jacobians is a single quick check.
- *The right notion of convergence for an instance-coupled estimator* is 8.2's: strong, target by target, never uniform. A frame fixed before the target is known has a blind unit direction. The adaptive order of choices is what turns strong convergence into recovery.
- *Continuous frame motion does not change rank.* The co-evolving frames of stage 16 are the analogue of the grading-preserving cocycle moves. Changing *what* is carried requires a separate, discrete regrading move. Its decoder complexity grows at least like $1/\varepsilon$, and on linear-response routes like the hard-edge law. This is a reading, not a theorem, for our commutative setting.